



Beni-Suef University – Faculty of Engineering

Department of Electronics & Communications Engineering | Second Year



Implementation of Cloud-Based Biometric Attendance System for Educators in a Developing Country

Published in Journal of Physics: Conference Series (2021)

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Paper Overview

Published in Journal of Physics: Conference Series · Vol. 2034 (2021) · DOI: 10.1088/1742-6596/2034/1/012018

Abstract

- ▶ Addresses limitations of paper-based attendance still dominant in developing countries
- ▶ Uses FPM10A fingerprint sensor for secure, tamper-proof authentication
- ▶ OLED display provides real-time user feedback during scanning
- ▶ ESP32 microcontroller uploads verified data to Google Drive via IFTTT webhooks
- ▶ Performance: authentication + cloud upload averages 2–5 seconds per candidate
- ▶ Free-tier cloud storage (15 GB) — zero annual maintenance cost

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Problem Statement

Why do we need a better attendance system?

Time & Accuracy

- ▶ Manual roll-call wastes valuable class time
- ▶ Human errors in recording are frequent
- ▶ Physical registers easily misplaced or stolen

Integrity Issues

- ▶ Proxy attendance (buddy punching) undetectable
- ▶ Signatures easily forged or signed for others
- ▶ No audit trail, no timestamp verification

Data Limitations

- ▶ No remote access for administrators
- ▶ Cannot auto-enforce exam eligibility criteria
- ▶ No structured data for performance correlation

"Despite numerous punitive measures, the traditional method has not proven effective for tracking punctuality in schools and at various workplaces over the years."

— Lasisi et al. (2021), Section 1

Limitations of Traditional Systems

Specific drawbacks documented in the paper and related literature



Paper-Based Register

- ▶ Time wasted calling names one by one in large classes
- ▶ Susceptible to impersonation and signature forgery
- ▶ Physical records lost, damaged, or stolen
- ▶ No real-time visibility for administrators or quality assurance
- ▶ Cannot automatically calculate attendance percentages



Prior Digital Attempts

- ▶ Expensive website infrastructure and regular maintenance [8]
- ▶ Requires multiple programming languages to operate
- ▶ Not portable — fixed to a single institutional location [27]
- ▶ RTC module needed for time sync (battery risk, clock drift)
- ▶ Integration costs prohibitive in developing-country contexts [9]



Design goal: Low-cost · Portable · Automated · Remote-accessible · No special skills required

Related Work — Biometric Authentication Taxonomy

Classification of biometric techniques per Thakur (2015) [1]



Physiological

Facial Recognition · Fingerprint · Iris Scan

Most documented category · Fingerprint: [7–11] · Face: [2–6] · Iris: [16–18]



Behavioural

Voice Pattern · Keystroke · Handwriting · Signature

Voice attendance: [12–15] · Effective but sensitive to environment noise



Chemical

Body Odour · Thermographs

Emerging category · Limited deployment documented in the literature

- ▶ Most prior systems are not commercially available, or too expensive for developing-country deployment.
- ▶ Selection rationale: Fingerprint chosen for cost-effectiveness, simplicity, and non-transferability (Section 2 of paper).

Why Fingerprint Biometrics?

Technical and contextual justification for the chosen modality



Uniqueness

Skin epidermis formed by hereditary + environmental factors — even identical twins differ [25]



Ease of Use

Single touch operation; no user training required; familiar interaction



Proven Technology

Deployed in ATMs, e-banking, airport check-in, access control doors [26]



Cost-Effective

Sensor modules affordable; no server infrastructure or website maintenance cost



Non-Transferable

Cannot be shared, lent, photocopied, or borrowed — eliminates proxy attendance



Developing-Country Fit

Works locally offline; only upload step needs internet — tolerates unstable connectivity

FPM10A Fingerprint Scanner — Specifications

Primary data acquisition component selected for its robust storage capacity

Technical Specifications

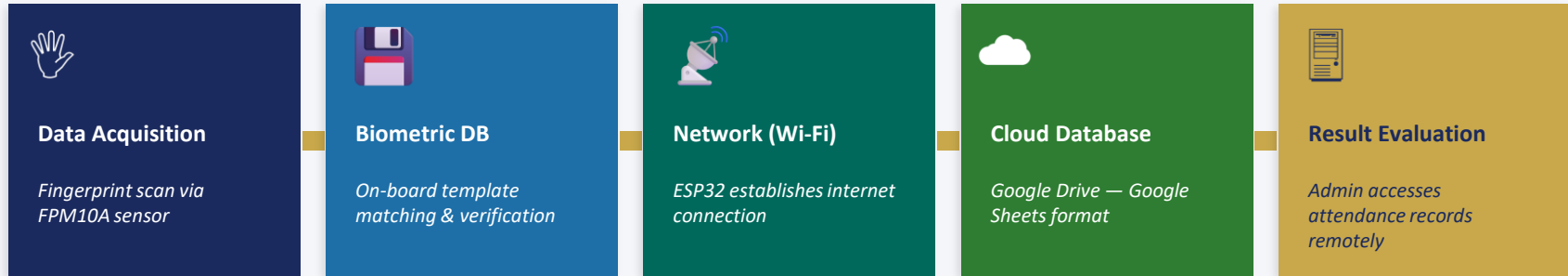
Sensor Type	Optical Fingerprint Scanner
Resolution	50 DPI (dots per inch)
Dimensions	55 mm × 23 mm × 21.5 mm
Interface	UART Serial (Tx / Rx)
Template Storage	On-board storage for enrolled fingerprints
Operating Voltage	3.3 V — supplied directly by ESP32
ESP32 Connection	UART2: Tx2 / Rx2 pins of ESP32-S

Role in the System

- ▶ Reads & captures user's fingerprint on touch
- ▶ Compares live scan against pre-stored templates on-board
- ▶ Returns match / no-match result to ESP32 via UART
- ▶ Chosen over LCD-based modules: OLED gives wider view angle and lower power
- ▶ Selected specifically for robust on-board storage capacity among available modules

System Architecture

End-to-end flow of the cloud-based biometric attendance system (Figure 1)



Only pre-registered candidates pass step 2 — system is exclusive by design. Authentication is local (fast); upload to cloud happens after match confirmed (network-dependent).

Hardware Components

Four integrated modules forming the physical prototype

FPM10A Fingerprint Scanner

- Reads & matches fingerprint
- UART interface: Tx2 / Rx2
- 50 DPI · 55×23×21.5 mm

0.96" OLED Display

- 128×64 pixel resolution
- Shows: boot, status, welcome/error
- Lower power & wider angle than LCD

ESP32-S Microcontroller

- 2.4 GHz Wi-Fi + Bluetooth combo
- TSMC 40 nm ultra-low-power chip
- Executes all logic & uploads data

Power Supply Unit

- Provides $V_{cc} = 3.3\text{ V}$ to all modules
- Common GND reference for circuit
- Simple regulated single-board supply

Integration

All modules share the 3.3 V rail from the power supply. D21/D22 (I²C) → OLED display. Tx2/Rx2 (UART2) → FPM10A sensor.

ESP32 Microcontroller — Central Intelligence

Espressif Systems · TSMC 40 nm ultra-low-power · Single 2.4 GHz Wi-Fi + Bluetooth combo chip [28]

Key Specifications

Wi-Fi	2.4 GHz 802.11 b/g/n
Bluetooth	BT 4.2 + BLE
Process Node	TSMC 40 nm ultra-low-power
GPIO Pins	34 programmable I/O pins
UART	3× UART (UART2 used for FPM10A: Tx2/Rx2)
I ² C	2× I ² C (used for OLED: D21=SDA, D22=SCL)
Operating Voltage	3.3 V
IDE	Arduino IDE (C/C++ firmware)

System Roles

- ▶ Executes all embedded firmware instructions
- ▶ Monitors data communication between all modules
- ▶ Establishes and manages Wi-Fi network connection
- ▶ Triggers IFTTT webhook on confirmed fingerprint match
- ▶ Sends HTTP POST request with attendance payload
- ▶ Controls OLED display status via I²C protocol
- ▶ Supplies 3.3 V power to FPM10A and OLED modules
- ▶ Programmed and flashed via Arduino IDE over USB

Circuit Design

Schematic overview and PCB layout of the prototype (Figures 2 & 3)

Power Input Terminal

- ▶ $V_{cc} = 3.3\text{ V}$ · GND common rail
- ▶ Single regulated supply for all modules

Fingerprint Sensor Interface

- ▶ FPM10A ↔ ESP32-S via UART2 (Tx2/Rx2)
- ▶ Bidirectional data · 3.3 V powered

OLED Display Interface

- ▶ ESP32 I²C: SCL→D22 · SDA→D21
- ▶ 128×64 px · 0.96" · 3.3 V supply

PCB Design Details (Figure 3)

- ▶ Top surface: ESP32-S board, OLED connector, fingerprint sensor header
- ▶ Bottom surface: routing traces, power rails, and ground plane for noise reduction
- ▶ Designed in electronic CAD (BiometricDB.pdsprj) — compact single-board form factor
- ▶ Firmware uploaded via USB using Arduino IDE — no external programmer required

Software Implementation — Arduino IDE Setup

Six-step configuration process (recreating Figure 4 of the paper)

1

OLED Linking & Programming

Adafruit SSD1306 library; boot logo, status bitmaps, fingerprint icons programmed

2

Board & Library Installation

ESP32 board package added to Arduino IDE; Adafruit FPM10A library installed

3

Mail Account Creation

Gmail account created to serve as IFTTT trigger/authentication account

4

Fingerprint Module Programming

FPM10A linked via UART2; enrollment sketch runs to store templates on-board

5

Cloud Storage Setup

IFTTT applet configured: webhook trigger → Google Sheets append action

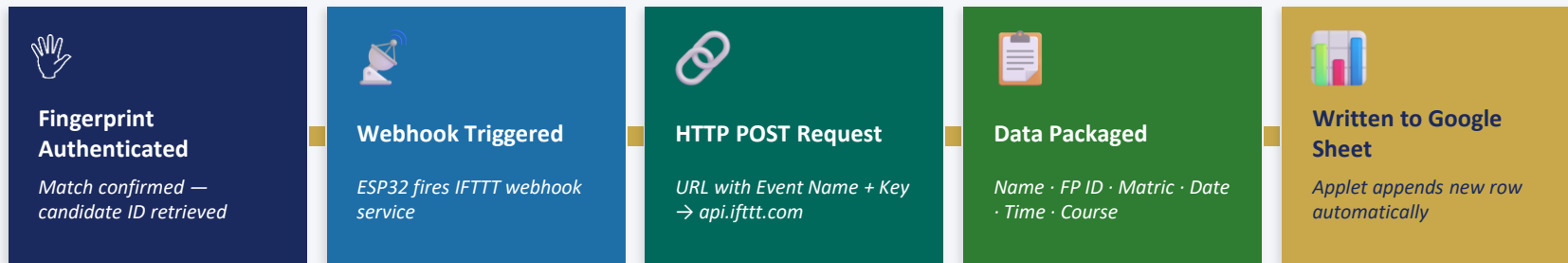
6

Applet Creation & Initiation

HTTP POST URL (event name + webhook key) embedded in ESP32 firmware

IFTTT Webhook Workflow

Automated pipeline from fingerprint match to Google Sheets entry



Webhook Details

- ▶ IFTTT ('If This Then That') automates the pipeline — zero custom server-side code required
- ▶ Webhook service fires each time a match is confirmed AND the Wi-Fi network is available
- ▶ HTTP POST sends JSON payload to IFTTT; applet action appends a row to Google Sheets
- ▶ Upload failure (poor network) → OLED shows crossed fingerprint; system resets gracefully

Cloud Integration & Data Storage

Google Drive (Google Sheets) as the free-tier attendance database

Date & Time	Course Code	Student Name	Fingerprint ID	Matric Number
Auto-captured from internet clock — no RTC module needed	IFTTT Event Name — enables filtering per course	First name of authenticated candidate	Unique numeric ID assigned during enrollment	University student identification number

Free Storage

15 GB Google Drive free tier — sufficient; no video or image files stored

Remote Access

Educator retrieves attendance sheet from any device, anywhere, anytime

Auto Timestamp

Internet clock provides date & time — eliminates battery-powered RTC module

QA Monitoring

Institution's quality assurance unit can monitor lecture compliance centrally

System Operation Flow

Runtime behaviour mapped to OLED display states (Figures 5, 6 & 7)

		OLED Display
1	Boot System starts; Adafruit logo displayed on OLED	<i>Boot Logo</i>
2	Wi-Fi Connect Scans for configured network; network name shown on OLED	<i>Connecting...</i>
3	Status Check OLED: 'Connected' ✓ or 'NotConnected' ✗ — decides next step	<i>Connected / Not</i>
4	Ready to Scan Fingerprint bitmap shown — system awaiting candidate finger	<i>Fingerprint Icon</i>
5	Scan & Match FPM10A reads live fingerprint; compares to stored templates	<i>Scanning...</i>
6	Auth Feedback Match → 'Welcome [Name]' / No match → 'Unregistered User'	<i>Match / Fail Msg</i>
7	Upload & Result Success → ✓ bitmap / Network fail → ✗ bitmap	<i>Upload Status</i>

Authentication Process

Fingerprint matching logic and identity verification flow

✓ MATCH FOUND

- ▶ Live fingerprint captured by FPM10A sensor
- ▶ Compared against pre-enrolled templates in on-board storage
- ▶ Match confirmed — candidate unique ID retrieved
- ▶ 'Welcome [Name]' displayed on OLED screen immediately
- ▶ Attendance management process initiated by ESP32
- ▶ Unique ID + metadata sent to cloud via IFTTT webhook
- ▶ Checked fingerprint bitmap displayed (upload confirmed)

✗ NO MATCH / UNREGISTERED

- ▶ Live scan does not match any stored template
- ▶ 'Unregistered User' displayed on OLED screen
- ▶ No attendance record created anywhere
- ▶ No data sent to cloud — clean rejection
- ▶ Crossed fingerprint bitmap displayed
- ▶ System resets and awaits next candidate

- ▶ Pre-Registration (one-time): fingerprint enrolled + unique ID assigned per candidate before deployment.
- ▶ Only pre-registered candidates can authenticate — the system is access-controlled by design.

Upload Process & OLED Feedback

HTTP POST payload structure and visual confirmation states

HTTP POST Payload — Data Fields

Field	Value Sent	Example
Student Name	First name (from enrolled template)	Collins
Fingerprint ID	Numeric ID assigned at enrollment	12
Matriculation No.	University student ID	EEE/2015/0003
Date	Auto from internet clock	January 19, 2021
Time	Auto from internet clock	09:00 PM
Course Code	IFTT Event Name	EEE 323

OLED Feedback States After Upload Attempt

- ▶  SUCCESS: Checked fingerprint bitmap → attendance confirmed and saved in Google Sheets
- ▶  FAILURE: Crossed fingerprint bitmap → poor network; record NOT saved; system resets
- ▶ Date & time auto-captured from internet clock — eliminates battery-powered RTC module entirely
- ▶ Upload failures observed during testing were resolved by switching to a more stable Wi-Fi network

Cloud Data Format — Google Sheets Output

Actual attendance data generated by the system (recreating Figure 8 from paper)

A — Date & Time	B — Course Code	C — Student Name	D — FP ID	E — Matric Number
Jan 19, 2021 09:00 PM	EEE 323	Collins	12	EEE/2015/0003
Jan 19, 2021 09:00 PM	EEE 323	Gloria	10	EEE/2015/0016
Jan 19, 2021 09:01 PM	EEE 323	Favour	2	EEE/2015/0077
Jan 22, 2021 06:04 PM	EEE 323	Joshua	12	EEE/2015/0023
Jan 27, 2021 11:53 PM	EEE 323	Pelumi	15	EEE/2015/0005
Jan 27, 2021 11:53 PM	EEE 323	Afeez	16	EEE/2015/0011

- ▶ Each row = one authenticated scan. Column A timestamp comes from internet (no RTC needed).
- ▶ Column B (Course Code) enables per-course filtering and automatic attendance calculation.
- ▶ Exportable directly for exam eligibility checks — the paper's primary use case.

Results & Performance

System testing outcomes and measured metrics



2 – 5 sec

Auth + Upload Time per Candidate



15 GB

Free Cloud Storage (Google Drive)



✓ 100%

Registered Candidates Correctly Identified



€ 0

Annual Maintenance Cost Required

Observed System Behaviour During Testing

- ▶ System boots and displays Adafruit logo on OLED; connects to Wi-Fi network automatically
- ▶ Registered fingerprints → 'Welcome [Name]' displayed; unregistered → 'Unregistered User'
- ▶ Successful upload → checked fingerprint bitmap; data appears in Google Sheets immediately
- ▶ Failed upload (poor network) → crossed fingerprint bitmap; record not saved — system stable
- ▶ Performance stabilised by switching to a more reliable Wi-Fi network — no hardware changes

System Advantages — Comparison with Prior Designs

Closing the gaps identified in related work (Refs [7–10, 27])

System	User Interface	Remote Access	Cost & Portability	Auth Method
Adewole et al. [7]	No UI	No remote access	Not scalable	✓ Fingerprint
Kadam et al. [8]	GUI + Website	✓ Online	High cost	✓ Fingerprint
Badejo et al. [9]	Portal UI	✓ Portal	Complex infra	✓ Fingerprint
Ahmed et al. [10]	LCD display	No cloud	LCD: high power	✓ Fingerprint
Narendra et al. [27]	✓ Cloud	✓ Cloud+local	Needs RTC, not portable	✓ Fingerprint
This Work (2021)	✓ OLED UI	✓ Google Drive	Portable · Free tier	✓ Fingerprint

✓ **This system uniquely combines: OLED UI · Free cloud storage · Portable design · No special skills · Zero annual cost**

Limitations, Future Work & Conclusion

Honest assessment of constraints, opportunities, and final takeaways

⚠️ Current Limitations

- ▶ Network-dependent upload: poor Wi-Fi causes unrecorded attendance
- ▶ Wet/worn fingers may affect fingerprint sensor read accuracy
- ▶ No offline buffering — failed records cannot be recovered later
- ▶ Prototype tested at small scale; large-scale deployment untested
- ▶ No fallback authentication if sensor hardware fails

👤 Future Work

- ▶ Local offline queue: store records; sync when connectivity restored
- ▶ Multi-modal fallback: fingerprint + RFID or PIN as backup
- ▶ Mobile app for real-time administrator notifications
- ▶ LMS/university portal integration for automated grade eligibility
- ▶ Multi-classroom simultaneous deployment and scale testing

✅ Conclusion

- ▶ Robust, economical, and user-friendly device delivered
- ▶ Zero special programming skills needed for operation or maintenance
- ▶ Free cloud tier (15 GB) sufficient for full academic-year records
- ▶ Saves educators' time; enforces punctuality culture in academia
- ▶ Enables automated exam eligibility enforcement from attendance data
- ▶ Supports QA monitoring of timetable compliance institution-wide



Thank You

Questions & Discussion

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